1) What this AI Agent is (scope, boundaries, deliverables)

Purpose (what it must achieve)

The agent is the **front gate** for RFQs/tenders: capture tender packages, convert them into **structured, review-ready intelligence**, and prep handover to procurement/engineering/management—**reducing manual hours but not replacing judgment**.

Non-negotiables (design constraints)

- **No system replacement** → integrate with existing email, storage, ERP, PM tools.
- **Human-in-the-loop always** (AI prepares → human approves) and **nothing is submitted automatically**.
- **Traceability over speed** → every output must point back to source doc + page/section.
- **Audit-ready by default** (logged, timestamped, reviewable).

Entry points & ingestion requirements

- Monitor tender inboxes + detect tender emails; generate Tender ID; store raw email metadata.
- Fetch attachments + linked docs (explicitly mentions **SharePoint / Drive / OneDrive**) and preserve original names + version history.
- Developer tickets already define **IMAP/Graph/Gmail** listener and **SharePoint/Drive/OneDrive link fetcher**, plus malware scanning.

Storage contract (super important for integrations)

Mandatory tender folder structure:
`/Tender_ID/01..08 (Instructions, Scope, Drawings, Specs, BOQ, Standards, Commercial, Output)`
Also: originals read-only + never overwrite + keep snapshots for addenda.

Core “intelligence” outputs

Your epics imply the agent produces (after OCR/classification):

- Tender instructions + deadlines + required docs + standards checklist
- BOQ parsing + scope/WBS + linking across scope/BOQ/drawings
- Compliance matrix + risks/gaps + assumptions + RFI drafts (human approved)
- “Tender Pack” export templates + **Procurement handover JSON**

Addendum/versioning flow (KSA-critical)

New email → match tender → detect addendum → delta compare → update impacted outputs → force re-approval.

---

2) Saudi construction: common third-party tools you should plan to integrate

Below is the **realistic integration set** you'll repeatedly hit with Saudi contractors + giga-projects + government work.

A) Government tenders platform (must-have for public sector)

Etimad (Ministry of Finance / Government Tenders Platform)

- Etimad provides services for browsing tenders and **submitting bids electronically**. (Etimad Portal)
  Integration implication: your agent should support:
- importing tender/RFP packs from Etimad (manual upload + assisted capture)
- storing Etimad tender IDs + deadlines + addendum notices
- generating Etimad-ready submission checklists (but still **no auto-submit**, per your scope)

B) Major owner / enterprise procurement portals (very common in KSA)

Saudi Aramco supplier ecosystem (SAP Ariba / e-Marketplace + Supplier Portal)
Aramco explicitly states suppliers manage interactions through systems including **SAP Ariba** as its e-Marketplace platform. (aramco.com)
Aramco also publishes a **SAP Ariba Suppliers Technical Guide**. (aramco.com)

Integration implication: support "supplier portal intake" patterns:

- RFQ invitations arrive via portal + email → agent ingests email + attachments + portal references
- map tender package → structured tender case → procurement handover JSON

C) Email + collaboration (almost always Microsoft in KSA enterprises)

Your spec already calls out **IMAP/Graph/Gmail** and link fetching from **SharePoint/Drive/OneDrive**.
In practice, Saudi enterprises heavily standardize on Microsoft 365, so Microsoft Graph + SharePoint/OneDrive connector is "day 1".

D) Document control / CDE (mega projects especially)

Autodesk BIM 360 / Autodesk Construction Cloud (ACC)
Autodesk University content shows **The Red Sea Development (TRSDC)** implemented **BIM 360** as the owner on large-scale projects. (Autodesk)
Integration implication: you need an ACC/BIM360 connector for:

- pulling tender drawings/specs/models

- writing back: compliance matrix, RFI drafts, submittal logs (usually via export + controlled upload)

Oracle Aconex (document control + audit trail)
Oracle positions Aconex around end-to-end collaboration with an **unalterable audit trail**—which matches your "audit-ready + traceability" principle. (Oracle)
For giga-project environments, Aconex is commonly specified; some NEOM-related procedures circulated online reference Aconex as the CDE (note: often hosted on third-party document sites, so treat as "confirm per client"). (Course Hero)

E) Planning & scheduling (very common for large EPCs)

Oracle Primavera P6 / Primavera Cloud
Aramco recruiting pages list experience in **Primavera P6** among planning software requirements (strong signal it's used in their environment). (careers.aramco.com)
Integration implication: import/export:

- WBS + tender milestone schedule + constraints
- risk flags tied to schedule impacts
  (Usually starts as Excel/XER/XML export; deeper APIs later.)

F) ERP / Finance backbone (common in big contractors)

SAP ERP (including SAP Cloud ERP / S/4 family)
SAP has a public case study of **Al-Rawaf Contracting Company** implementing **SAP Cloud ERP**. (SAP)
Integration implication: your tender intelligence should map cleanly into ERP objects:

- project/job setup, cost codes, BOQ line mapping, vendor package structure, approvals evidence

(Also common in region: Oracle ERP/Fusion; you'll often see SAP + Oracle mixes, but I'm only stating what we have solid sources for here.)

G) Construction management platforms (more common with international contractors, increasing in region)

Procore
Procore expanded into **MENA** and emphasizes integrations via its ecosystem marketplace. (Procore)
Integration implication: treat Procore as "Phase 2/3 connector" depending on your target client base (international JV contractors use it more often).

---

3) Recommended integration roadmap (aligned to your scope)

MVP (fastest path; matches your PDF exactly)

1. **Email ingestion** (Graph + IMAP + Gmail)
2. **Storage connectors**: SharePoint/OneDrive + Google Drive + S3/Azure Blob
3. **Folder structure + versioning snapshots + audit log**
4. **Export pack + procurement handover JSON**

Phase 2 (Saudi-specific deal-closers)

- **Etimad intake support** (guided import, tender metadata capture, addendum tracking) (Etimad Portal)
- **Aramco SAP Ariba / e-Marketplace intake patterns** (aramco.com)
- **ACC/BIM360 connector** (especially if targeting Red Sea / PIF ecosystem contractors) (Autodesk)

Phase 3 (enterprise depth)

- **Oracle Aconex connector** (document control workflows + transmittals) (Oracle)
- **Primavera** import/export pipeline (careers.aramco.com)
- **SAP ERP integration** (project/job + cost codes + approvals evidence) (SAP)
- Optional: **Procore** connector for certain contractors/JVs (Procore)

---

4) Integration "contract" (what data we must map consistently)

To avoid "custom integration chaos", standardize these objects internally (then map outward):

- **TenderCase** (Tender ID, client/owner, deadlines, portal refs like Etimad/Ariba)
- **Document** (hash, version, source link, classification, page-level citations) — required by traceability
- **BOQItem** (item no, description, qty, unit, discipline, linked drawing refs)
- **ComplianceItem** (requirement, evidence link, status, reviewer approval)
- **RFI / Assumption / Risk** (each linked to exact source sections; human-approved)
- **HandoverPayload** (attachments map + JSON schema to procurement agent)

---

1) Goal and expected outcome

Goal

Build an AI-assisted Tender/RFQ intake + analysis system that:

- Ingests tender invitations (email + portal links + uploaded files)
- Organizes documents into a strict folder structure
- Extracts key tender requirements (deadlines, compliance, scope, BOQ, risks, RFIs)
- Produces an **audit-ready tender pack** and a **handover payload** for Procurement/Estimating
- Keeps a **human-in-the-loop** approval step for every final output (no auto-submissions)

What "done" looks like

For any tender invitation, the system can produce:

- A Tender Case with unique ID + metadata
- Clean document library + version history (including addenda)
- Tender summary + compliance matrix + risk/assumptions list + RFI draft list
- BOQ structured output (CSV/JSON) where applicable
- Export pack (PDF/Docx/Excel templates) + a standard "handover.json"

---

2) Scope boundaries (non-negotiables)

- **Human approval required** before any final deliverable is marked "Ready"
- **Never overwrite originals** (read-only originals + versioned snapshots)
- **Everything must be traceable**: outputs link back to source file + page/section
- No "submit bid" automation (we support preparation only)
- Must be secure-by-design (malware scan, RBAC, audit logs)

---

3) System architecture (high-level)

Components

1. **Ingestion Layer**
   - Email listener(s): Microsoft Graph (primary), IMAP fallback, Gmail support
   - Link fetcher: SharePoint / OneDrive / Google Drive links
   - Manual upload portal (drag/drop tender packs)
2. **Storage Layer**
   - Primary: SharePoint/OneDrive (common in KSA enterprises)
   - Optional: S3 / Azure Blob (enterprise setups)
   - All stored under Tender ID folder structure
3. **Processing Layer**
   - Document classification: instructions/specs/drawings/BOQ/commercial/standards

    - OCR for scanned PDFs
    - Extraction pipelines:
        - Tender rules & deadlines
        - Compliance matrix
        - BOQ parsing
        - Risk/assumptions
        - RFI draft generation
    - Addendum delta comparison + re-approval triggers
4. **App Layer**
    - Tender dashboard (status, deadlines, outputs, approvals)
    - Reviewer workflow (procurement/engineering/commercial)
    - Export + handover generation
5. **Integration Layer**
    - Saudi-specific external tools (Etimad, SAP Ariba patterns, ACC/BIM360, Aconex etc.)
    - ERP/PM exports (Primavera, SAP)

---

4) Folder structure standard (must implement exactly)

For every Tender Case, create:

```
/Tender_<ID>/
```

- `01_Instructions`
- `02_Scope`
- `03_Drawings`
- `04_Specifications`
- `05_BOQ`
- `06_Standards`
- `07_Commercial`
- `08_Output` (generated deliverables only)

Rules:

- Store email + attachments + fetched linked docs (with original file names)
- All originals read-only + hash stored (SHA256)
- Addenda stored as new versions; never replace prior docs

---

5) Data model (minimum objects)

Core entities

- **TenderCase**
    - tender_id, title, owner/client, source (email/portal/manual), due_date/timezone, status

    - portal_refs (Etimad ID, Ariba ref, etc.)
- **Document**
    - tender_id, doc_id, filename, type, version, source_url, hash, created_at
- **ExtractedItem**
    - type: deadline / requirement / compliance / risk / assumption / RFI / BOQ line
    - value fields + **source citation** (doc_id, page, snippet)
- **Approval**
    - artifact_id, approver, status, timestamp, comments
- **AuditLog**
    - who did what, when, from where, on which object

Output artifacts

- tender_summary.md/pdf
- compliance_matrix.xlsx
- boq_structured.csv/json
- risks_assumptions.xlsx
- rfi_drafts.docx
- handover.json

---

6) Core workflows to implement

A) New Tender Intake (Email / Manual / Link)

1. Detect tender email or manual upload
2. Create TenderCase + Tender ID
3. Download/store all files (attachments + linked docs)
4. Classify docs into 01–08 folders
5. Run malware scan + OCR if needed
6. Generate first-pass outputs (Summary, Compliance, Risks, BOQ)
7. Set status: **“Needs Review”**

B) Review & Approval

1. Reviewer sees outputs + citations
2. Reviewer can:
    - Accept / edit / reject items
    - Add missing requirements
    - Approve deliverables
3. Once all mandatory approvals complete → status: **“Approved / Ready for Handover”**
4. Export pack created in `08_Output`

C) Addendum Handling (must be robust)

1. New email arrives referencing existing tender OR manual addendum upload
2. Save addendum as new version(s)

3. Run delta compare:
    - deadlines changed?
    - scope/BOQ changed?
    - commercial terms changed?
4. Mark impacted artifacts as **“Re-approval Required”**
5. Notify relevant reviewers

---

7) Integration roadmap (Saudi-focused)

MVP (build first)

- Microsoft 365 Graph (email + attachments)
- IMAP fallback + Gmail support
- SharePoint/OneDrive link fetching
- Google Drive link fetching
- Storage support: SharePoint/OneDrive + optional S3/Azure Blob
- Export pack + handover.json

Phase 2 (Saudi high-value)

- **Etimad support** (government tenders)
    - Tender metadata capture (Etimad tender ID, deadlines, addendum notices)
    - Assisted import (manual upload + structured fields)
- **SAP Ariba intake patterns**
    - Most common: portal sends email notifications + attachments/links
    - We support capture of Ariba reference IDs + file intake
- **Autodesk ACC/BIM360 connector (optional for mega projects)**
    - Pull drawings/specs; export outputs back as controlled uploads

Phase 3 (enterprise depth)

- **Oracle Aconex (document control)** connector (download + controlled upload)
- **Primavera P6 integration** (import/export schedules; milestone mapping)
- **SAP ERP integration** (export tender handover to cost codes/project setup)
- Optional: Procore connector (JV/international contractors)

---

8) Security + compliance requirements

- Malware scanning on every file ingested
- RBAC:
    - Super Admin
    - Tender Admin
    - Reviewer (Procurement/Commercial/Engineering)
    - Viewer
- Audit log for all actions

- Encryption at rest + TLS in transit
- Data residency option (Saudi clients may require regional hosting)
- “No hallucination” rule: every extracted requirement must carry a source citation or be flagged “Needs Human Input”

---

9) Engineering tasks (assignable checklist)

Backend (API + processing)

- [ ] TenderCase CRUD + status machine
- [ ] Email ingestion (Graph) + IMAP/Gmail fallback
- [ ] Link fetchers (SharePoint/OneDrive/Drive)
- [ ] File pipeline: store → hash → malware scan → OCR → classify
- [ ] Extraction services:
  - [ ] Instructions/deadlines extractor
  - [ ] Compliance matrix generator
  - [ ] BOQ parser (tables → structured output)
  - [ ] Risks/assumptions/RFI generator
- [ ] Addendum delta compare + artifact invalidation
- [ ] Export engine (docx/xlsx/pdf)
- [ ] Handover JSON schema + generator
- [ ] Audit log + RBAC enforcement

Frontend (dashboard + approvals)

- [ ] Tender list + filters + SLA countdown to due date
- [ ] Tender details: docs, versions, classifications
- [ ] Output viewer with citations (click → open source page)
- [ ] Reviewer approval workflow (approve/reject/comment)
- [ ] Notifications (re-approval required)

DevOps

- [ ] Secrets management (Graph creds, storage creds)
- [ ] Logging/monitoring + alerting
- [ ] Backup strategy + retention policy
- [ ] Environment separation (dev/staging/prod)

10) Definition of Done (MVP acceptance criteria)

A tender email with attachments + SharePoint links should result in:

- A TenderCase created with correct folder structure
- All documents stored, hashed, malware-scanned, OCR’d when needed
- Auto-generated: summary, compliance matrix, risks/assumptions, and handover.json
- Reviewer can approve outputs; system locks final versions into `08_Output`
- Addendum triggers delta compare and forces re-approval
- Full audit log exists for all actions